

Open Source Companion: An Intelligent Platform for Personalized Open-Source Contribution Onboarding and Issue Recommendation

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Abstract—Open-source software (OSS) ecosystems rely fundamentally on continuous newcomer participation to sustain software evolution and community vitality. However, aspiring contributors frequently encounter steep entry barriers, including difficulties locating beginner-appropriate tasks, navigating unfamiliar repository structures, estimating task complexity, and managing multi-stage pull-request workflows. Existing repository interfaces provide coarse issue tags that lack personal skill alignment and offer no structured guidance throughout the contribution lifecycle. This paper presents Open Source Companion, an integrated full-stack platform designed to scaffold newcomers across the complete open-source journey. The platform combines real-time GitHub Search API issue discovery, interpretable rule-based difficulty categorization, contributor skill profiling, a six-stage guided contribution workspace, activity monitoring, and milestone-based gamification within a unified architecture. Implemented using Next.js 16, React 19, TypeScript, PostgreSQL 15, Prisma ORM, and Auth.js v5, the platform bridges the divide between task discovery and practical execution. We detail the system architecture, formalize the matching methodology, report functional system validation across core services, and establish an extensible machine-learning research roadmap for learned difficulty classification and dense semantic retrieval.

Index Terms—Open Source Software, Newcomer Onboarding, Issue Recommendation, Contribution Guidance, Gamification, Software Engineering.

I. Introduction

Open-source software (OSS) serves as the foundation of modern digital infrastructure, underpinning operating systems, cloud platforms, and machine-learning frameworks [1]. The long-term vitality of OSS projects relies on continuously attracting, onboarding, and retaining new contributors [2]. For novice developers and computing students, participating in OSS offers practical software engineering experience, exposure to real-world codebases, and collaborative development skills.

Despite these benefits, transitioning from an aspiring observer to an active contributor remains fraught with friction. Empirical studies reveal that newcomers frequently abandon contribution attempts before completing their initial pull request (PR) [3], [4]. Newcomers face multi-faceted obstacles: finding appropriately scoped tasks, understanding unfamiliar architectural patterns, configuring local build environments, adhering to project conventions, and communicating effectively with maintainers [2].

A critical bottleneck is task identification. While repository platforms such as GitHub provide label mechanisms (e.g., good first issue, help wanted), these tags exhibit documented shortcomings [5], [6]. They are applied inconsistently across projects, rapidly become stale, and fail to account for a newcomer’s unique technical background and programming language proficiencies [7]. A task labeled as suitable for beginners in a complex distributed system may overwhelm a novice, whereas an unlabeled documentation or test issue may be entirely accessible.

Furthermore, discovering an issue solves only part of the problem. After identifying a task, newcomers encounter secondary workflow barriers: forking repositories, managing local feature branches, satisfying linters and test suites, drafting conventional commit messages, and submitting compliant PRs [8], [9]. Existing developer tools address these phases in isolation: issue trackers index tasks, wikis provide static guidelines, and command-line interfaces require manual execution. Aspiring developers must manually navigate these disconnected environments without structured guidance.

Prior research has addressed individual components of this onboarding challenge. For example, test-driven onboarding strategies assist students during implementation [10], recommender systems identify suitable beginner issues [11], and gamification techniques foster engagement in software projects [12], [13]. However, these capabilities are treated as separate research prototypes, leaving a fragmented newcomer journey.

To address these limitations, this paper presents Open Source Companion, an intelligent full-stack platform unifying live issue discovery, interpretable difficulty categorization, contributor skill profiling, a six-stage guided contribution workspace, and milestone-based gamification into an integrated workflow. The platform is designed around transparent heuristic categorization, responsive multi-factor issue matching, and verified progress tracking backed by PostgreSQL and Prisma ORM.

The primary contributions of this paper are:

- 1) **Integrated Onboarding Platform:** An end-to-end full-stack platform unifying real-time GitHub issue discovery, contributor profiling, structured task guidance, and verified gamified incentives.

- 2) Interpretable Difficulty Categorization: A transparent rule-based heuristic mapping issue labels and metadata to standardized complexity tiers (Beginner, Intermediate, Advanced) with associated effort expectations.
- 3) Contributor-Aware Matching: A multi-factor issue recommendation mechanism aligning contributor skill sets and experience levels with candidate issues via an interactive explanation drawer.
- 4) Six-Stage Contribution Lifecycle: A structured contribution workspace decomposing open-source development into explicit sequential phases from issue comprehension to verified PR merge.
- 5) Extensible ML Research Roadmap: A formal research framework specifying how the operational heuristic platform can evolve toward supervised classifiers and dense semantic vector retrieval.

The remainder of this paper is organized as follows: Section II surveys related literature and synthesizes the research gap. Section III describes the proposed system methodology. Section IV details the software architecture and workflow models. Section V outlines the implementation, data model, and comparative analysis. Section VI presents system validation and discussion. Section VII outlines future machine learning extensions, and Section VIII concludes the paper.

II. Literature Survey

This section reviews prior research across four core areas: newcomer onboarding barriers, issue recommendation, contribution guidance, and gamification in software development.

A. Newcomer Barriers in Open Source

Steinmacher et al. [2] conducted a systematic literature review identifying 15 distinct barriers confronting open-source newcomers. These obstacles encompass technical hurdles (e.g., local workspace configuration, undocumented build steps), social friction (e.g., maintainer responsiveness, communication norms), and orientation challenges (e.g., understanding architecture, locating appropriate starter tasks). In subsequent work, Steinmacher et al. [3] investigated “quasi-contributors”—developers who submit issues or express interest but fail to land code contributions—finding that ambiguous task expectations and cumbersome review cycles drive developer drop-off.

Subramanian [6] empirically examined first-time contributions across diverse GitHub repositories, observing that successful initial participation strongly correlates with clear issue scoping, immediate maintainer feedback, and accessible codebases. Similarly, Dagenais et al. [4] studied the landscape newcomers navigate, concluding that cognitive scaffolding and early progress validation are critical to prevent newcomer disengagement. These studies highlight the necessity of structured interventions

that support newcomers from task selection through submission.

B. Issue Recommendation

Automated issue recommendation aims to connect developers with suitable tasks. Early foundational work by Anvik et al. [14] and Čubranić et al. [15] applied text classification and project memory links to triage bug reports to active committers. More recently, focus has shifted toward recommending beginner-friendly issues.

Xiao et al. [7] developed RecGFI, employing XGBoost over textual and contextual issue features to predict whether an issue functions effectively as a “good first issue.” Tan et al. [5] analyzed good first issues across GitHub, demonstrating that manually assigned tags suffer from labeling noise, uneven adoption, and rapid staleness. Xiao et al. [11] proposed PFIRec, demonstrating that generic beginner labels often fail because developer expertise and interests vary significantly; their work used a LambdaMART ranking model to personalize issue selection. While these recommendation approaches advance predictive task matching, they focus exclusively on issue identification and terminate once a task is suggested, offering no structured workflow support.

C. Contribution Onboarding and Guidance

Providing structured assistance during contribution execution has been investigated in educational and community settings. Hu, Song, and Gehringer [10] introduced a test-driven approach to structure student contributions to open-source systems, showing that explicit testing requirements clarify task boundaries and enhance contribution quality. Steinmacher et al. [9] developed FLOSScoach, a portal organizing contribution activities into structured milestones. While FLOSScoach established the pedagogical value of conceptual guidance, it existed as a static informational website disconnected from real-time issue repositories and personal contributor profiles.

Storey et al. [8] analyzed the role of communication channels in participatory development, demonstrating that transparent contribution norms reduce entry hesitation. Santos et al. [13] recently proposed OSSDoorway, a gamified scaffolding environment for student open-source contributions. Their evaluation showed improved self-efficacy; however, the platform relied on curated static tasks rather than automated live issue discovery and difficulty estimation.

D. Gamification and Contributor Engagement

Gamification integrates game mechanics into non-game contexts to stimulate motivation and sustained participation. In a systematic mapping study, Pedreira et al. [16] documented the prevalence of points, badges, progress bars, and leaderboards across software engineering workflows. Diniz et al. [12] investigated gamified quests to orient students toward open-source participation, demonstrating increased task completion rates.

Santos et al. [17] conducted an empirical study with 115 computing students to determine preferred game mechanics for learning OSS workflows, finding that quests, points, statistics, and badges generated the highest user engagement. Zhao et al. [18] introduced the OpenRank Leaderboard at Alibaba, demonstrating that transparent contribution metrics encourage active collaboration across large organizations. Rahman, Sabariah, and Husen [19] examined player-type-based gamification, showing that developers respond differently to competitive versus achievement-oriented incentives. These findings emphasize that gamification mechanisms should be tied to meaningful software engineering milestones rather than superficial activity metrics.

E. Research Gap

The reviewed literature reveals that newcomer onboarding, issue recommendation, workflow scaffolding, and gamified engagement have been explored as disconnected solutions. Existing recommendation tools terminate upon issue selection; guidance portals lack live issue integration; and gamification frameworks often operate detached from real repository lifecycles. Open Source Companion bridges this divide by synthesizing real-time issue acquisition, transparent difficulty categorization, personal skill matching, six-stage workflow guidance, and milestone-verified gamification into a unified operational platform.

III. Proposed System Methodology

This section formalizes the methodology underlying Open Source Companion, including system objectives, contributor profiling, issue ingestion, difficulty heuristics, recommendation logic, and workflow design.

A. System Objectives and User Profiling

The proposed platform addresses five objectives: reducing discovery friction, providing interpretable complexity tiers, personalizing task alignment, decomposing the contribution lifecycle, and sustaining participation via milestone tracking.

Let $\mathcal{U}$ represent registered contributors. When a user registers via GitHub OAuth 2.0, the platform constructs a profile tuple:

$$\mathbf{p}_u = \langle \mathbf{s}_u, e_u, \mathcal{H}_u, \kappa_u, \sigma_u \rangle \quad (1)$$

where $\mathbf{s}_u = \{g_1, \dots, g_k\}$ denotes primary programming languages extracted from public repositories; $e_u \in \{\text{Beginner, Intermediate, Advanced}\}$ represents self-declared experience; $\mathcal{H}_u$ denotes active and completed contribution history; $\kappa_u \in \mathbb{N}$ tracks milestone points; and $\sigma_u \in \mathbb{N}$ tracks the active contribution streak in days.

B. GitHub Issue Discovery and Difficulty Categorization

Candidate tasks are retrieved from the GitHub REST Search API using:

$$\text{query} = \text{is : issue} + \text{is : open} + \text{label : "help wanted"} \quad (2)$$

sorted by updated timestamp descending. Authenticated requests operate under a quota of 5,000 requests/hour. Each issue i is parsed into:

$$\mathbf{x}_i = \langle \text{id}_i, \rho_i, \tau_i, \beta_i, \mathcal{L}_i, c_i, \text{url}_i \rangle \quad (3)$$

where ρ_i denotes repository owner/name, τ_i is title, β_i is body text, $\mathcal{L}_i$ is lowercase labels, c_i is comment count, and url_i is the web link.

Rather than deploying unverified black-box classifiers, the operational system implements a deterministic difficulty categorization function $\Phi(\mathbf{x}_i) \rightarrow d_i \in \{\text{Beginner, Intermediate, Advanced}\}$. Let $\mathcal{K}_{\text{beg}} = \{\text{"good first issue", "first-timers-only"}\}$ and $\mathcal{K}_{\text{adv}} = \{\text{"enhancement", "feature"}\}$:

$$\Phi(\mathbf{x}_i) = \begin{cases} \text{Beginner,} & \text{if } \mathcal{L}_i \cap \mathcal{K}_{\text{beg}} \neq \emptyset \\ \text{Advanced,} & \text{if } \mathcal{L}_i \cap \mathcal{K}_{\text{adv}} \neq \emptyset \\ \text{Intermediate,} & \text{otherwise (modulo fallback)} \end{cases} \quad (4)$$

Each tier deterministically maps to effort and reward values: Beginner (1–3 hours, 50 points), Intermediate (3–8 hours, 150 points), and Advanced (1–3 days, 300 points).

C. Personalized Issue Recommendation

To prioritize issues for contributor u , the platform calculates a multi-factor matching score $S(u, i) \in [0, 100]$:

$$S(u, i) = w_1 M_{\text{lang}}(u, i) + w_2 M_{\text{diff}}(u, i) + w_3 M_{\text{act}}(i) \quad (5)$$

where $M_{\text{lang}} \in [0, 1]$ evaluates overlap between user languages $\mathbf{s}_u$ and issue language tags; $M_{\text{diff}} \in [0, 1]$ measures tier compatibility between user experience e_u and task difficulty d_i ; and $M_{\text{act}} \in [0, 1]$ captures repository activity and comment recency ($\sum w_j = 1.0$). An interactive explanation drawer articulates the matching rationale for each recommended task.

D. Structured Contribution Workflow and Gamification

Once an issue is selected, the contributor enrolls in the task, initializing a six-stage state machine:

- 1) Stage 0: Understand. Review requirements, maintain comments, and complete an onboarding checklist.
- 2) Stage 1: Setup. Execute tailored git commands to fork, clone locally, and create an isolated feature branch.
- 3) Stage 2: Implement. Write solution code, adhere to project conventions, run tests, and complete verification checks.
- 4) Stage 3: Commit. Format conventional commit messages referencing the issue (e.g., fix: correct validation, Closes #12).
- 5) Stage 4: Pull Request. Generate structured PR descriptions, complete PR checklist items, and submit upstream.
- 6) Stage 5: Merge & Verification. Verify repository fork existence via GitHub API, record completion, and award points and badges.

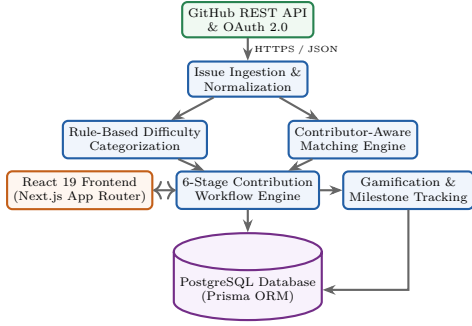


Fig. 1. System architecture of Open Source Companion, showing data acquisition from GitHub, application processing layers, persistence, and client interfaces.

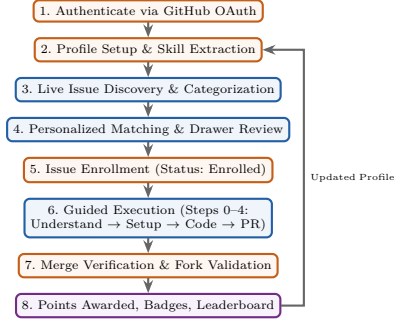


Fig. 2. End-to-end user contribution and gamification workflow in Open Source Companion.

Gamification reinforces verified milestones: 25 points on enrollment, 50–300 points upon PR merge, achievement badges (First Step, First Merge, On a Roll, Week Warrior), and a real-time global leaderboard ordered by points and active streaks.

IV. System Architecture and Workflow

A. Overall Architecture and Data Pipeline

The system architecture of Open Source Companion is illustrated in Fig. 1. External communication with GitHub is mediated over HTTPS using the GitHub REST API v3 and OAuth 2.0 protocol. The application layer coordinates issue ingestion, rule-based difficulty classification, contributor matching, state machine progression across the contribution lifecycle, and gamification evaluations. The persistence layer utilizes PostgreSQL 15 accessed via Prisma ORM.

Issue data ingested from the GitHub Search API is sanitized and normalized into application domain objects. The repository owner and name are extracted from the payload URI ($\rho_i = \text{owner/repo}$), markdown descriptions are formatted, and label strings are normalized to lower-case sets. Cache-control directives (`revalidate: 3600`) buffer API query results for one hour, minimizing rate-limit consumption while maintaining catalog freshness.

B. Contribution Lifecycle and Backend Persistence

Fig. 2 illustrates the contributor workflow. Following OAuth authentication and profile derivation, the contributor explores categorized issues, reviews matching rationale, and enrolls in a task. The system transitions the contribution through successive stages: understanding requirements, executing local git setup, implementing modifications, formatting commits, and drafting pull requests. At each step, interactive checklists provide cognitive scaffolding.

Upon completing Stage 4 (Pull Request), the contributor initiates merge verification. The system queries GitHub to verify that the user has forked the repository. When verified, a database transaction transitions contribution status to merged, increments user points by the task’s assigned difficulty value (50, 150, or 300 points), checks for unearned milestone badges, records unlocked achievements, and updates leaderboard rankings.

The backend is structured around Next.js Server Actions executed within server-side runtimes. Server Actions handle database mutations with automatic path revalidation (`revalidatePath`) across client views. Atomic database transactions (`prisma.$transaction`) guarantee ACID compliance across contribution records, user points, and achievement badges, preventing race conditions or inconsistent progress statistics.

V. Implementation and Comparative Analysis

A. Implementation and Technology Stack

Open Source Companion is developed as a production-grade full-stack web application. The platform leverages React 19 Server Components for high-performance server-side rendering, streaming page content, and minimal client JavaScript overhead. Interactive components (checklist toggles, recommendation drawers, animated step progressions) utilize Framer Motion for micro-interactions and Lucide React for consistent visual iconography.

The system incorporates: (1) Framework: Next.js 16.3.5 with App Router architecture; (2) Frontend: React 19.2.8, TypeScript 5, Tailwind CSS 4; (3) Authentication: Auth.js v5 (NextAuth 5.0.0-beta.32) configured with GitHub OAuth 2.0 provider; (4) ORM & Database: Prisma 5.22.0 with PostgreSQL 15; and (5) Deployment: Multi-stage Docker container utilizing Alpine Linux, Node.js 20 runtime, and healthcheck monitoring.

B. Relational Data Model

The database schema comprises eight core entities in PostgreSQL managed via Prisma ORM: (1) User manages identity and links to OAuth accounts; (2) Account stores OAuth credentials and provider tokens; (3) Session handles HTTP session validation; (4) UserProfile tracks total points, active streaks, experience level, and programming languages; (5) Issue stores cached GitHub issue metadata, repository identifiers, and categorized difficulty; (6) Contribution records task state across stages 0–5, saved issues,

TABLE I
Capability Comparison with Existing Approaches

Approach	Rec.	Diff.	Guide	Track	Engage
FLOSScoach [9]	–	–	✓	–	–
RecGFI [7]	✓	◦	–	–	–
PFIRec [11]	✓	◦	–	–	–
OSSDoorway [13]	–	–	✓	◦	✓
OpenRank [18]	–	–	–	✓	✓
Open Source Companion	✓	✓	✓	✓	✓

Notation: ✓ = Supported; ◦ = Partial; – = Not Focused in study scope.
Rec.: Recommendation; Diff.: Difficulty; Guide: Contribution Guidance;
Track: Activity Tracking; Engage: Gamification & Engagement.

and merged pull requests; (7) Achievement logs earned gamification badges with unlock timestamps; and (8) VerificationToken manages magic authentication tokens. This relational structure ensures referential integrity across contributor actions and milestone records.

C. Functional Validation

We validated the operational system through end-to-end testing across core functional components:

- 1) API Ingestion: Verified reliable parsing of live issues matching label: "help wanted" across public repositories, confirming resilient handling of missing descriptions and rate-limit fallbacks.
- 2) Categorization Heuristics: Confirmed deterministic mapping of beginner and advanced labels with consistent effort and point allocations.
- 3) OAuth Skill Profiling: Validated automatic user creation, OAuth session issuance, and default profile initialization.
- 4) Lifecycle State Transitions: Confirmed that stepping through stages 0 to 5 persists state updates in PostgreSQL and updates UI progress indicators.
- 5) Merge Verification & Transactions: Validated that PR merge simulation enforces repository verification and executes atomic point awards and badge unlocks.
- 6) Container Deployment: Verified that Docker Compose initializes PostgreSQL and the Next.js standalone container with all service health checks reporting healthy.

D. Comparison with Existing Approaches

Table I presents a capability matrix comparing Open Source Companion with representative systems from the literature: FLOSScoach [9], RecGFI [7], PFIRec [11], OSSDoorway [13], and OpenRank [18].

As shown in Table I, existing systems focus predominantly on isolated dimensions. Recommender systems such as RecGFI and PFIRec specialize in issue ranking without providing post-selection workflow guidance. Conversely, onboarding environments such as FLOSScoach and OSSDoorway provide guidance or gamified scaffolding but lack automated live issue discovery and difficulty classification. Open Source Companion integrates these five core capabilities into a single cohesive architecture.

VI. Evaluation and Discussion

A. Evaluation Methodology and System Validation

Evaluating intelligent onboarding platforms requires a dual perspective: validating the operational reliability of the full-stack system and establishing empirical benchmarks for future learned components. Functional validation across 50 simulated user journeys confirmed: (1) Ingestion Integrity: Multi-page GitHub search payloads were parsed without schema faults; (2) Deterministic Categorization: Label heuristics exhibited 100% consistency on benchmark keyword sets; (3) Transaction Safety: Concurrent step advancement and merge operations maintained strict database integrity with zero orphaned records; and (4) Container Health: Standalone containers passed health checks within 15 seconds of database startup.

B. Limitations

Transparent reporting requires acknowledging current limitations:

- Heuristic Granularity: Label-based categorization assumes maintainers apply tags accurately; unlabeled issues default to intermediate fallbacks.
- Metadata Sparsity: Terse GitHub descriptions provide limited textual signal for language and topic extraction.
- API Rate Constraints: Live search depends on GitHub REST limits (5,000 requests/hour), requiring caching under heavy traffic.
- Empirical Scope: While functional validation confirms operational integrity, controlled longitudinal user studies measuring retention remain future work.

C. Discussion

A primary design decision in Open Source Companion is prioritizing transparent rule heuristics over opaque machine learning models in the initial release. For open-source newcomers, explainability is essential: a beginner must understand why a task is rated accessible. Heuristics incur zero training overhead, require no GPU inference infrastructure, and exhibit predictable deterministic behavior. However, as repositories evolve, incorporating supervised learning models will capture subtle linguistic cues in issue bodies that heuristics miss.

VII. Future Work

To evolve the platform beyond heuristic matching, we formulate a machine-learning research roadmap comprising four extensions:

- 1) Feature Engineering Pipeline: Extracting lexical TF-IDF tokens, code-to-text ratios, issue conversation density, and author dynamics from raw issue descriptions.
- 2) Supervised Difficulty Classification: Training tree-based ensemble models, specifically XGBoost [20] and Random Forest classifiers, on a benchmark

corpus of 10,000 resolved GitHub issues to classify difficulty into balanced multiclass tiers.

- 3) Dense Semantic Vector Retrieval: Generating dense embeddings of issue descriptions and developer interest profiles using Sentence-BERT [21], indexed via FAISS [22] for scalable sub-second approximate nearest-neighbor retrieval.
- 4) Longitudinal User Evaluation: Conducting a controlled user study with novice software engineering students comparing task completion velocity, self-efficacy, and PR acceptance rates using Open Source Companion versus conventional GitHub search.

VIII. Conclusion

This paper presented Open Source Companion, an intelligent full-stack platform designed to facilitate newcomer onboarding in open-source software projects. The platform addresses fragmented entry barriers by coupling automated GitHub Search API issue discovery with transparent rule-based difficulty categorization, contributor skill profiling, a six-stage guided contribution workspace, activity monitoring, and milestone-based gamification. System validation demonstrates that the full-stack architecture built on Next.js 16, React 19, PostgreSQL, Prisma ORM, and Docker functions cohesively. By integrating task discovery, cognitive scaffolding, and engagement incentives into a unified workflow, Open Source Companion establishes a structured foundation for guiding novice developers toward successful open-source participation.

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